ECS 240 Project Report

Rajiv Choudhury, Jacob Hamzawi, Rishika Nigam, Gurpreet Singh

November 13, 2026

1 Complete Formal Grammar (In Progress)

$$
\begin{aligned}
S &::= \\
\texttt{<Int>} &::= 0 \mid \texttt{<NonZeroInt>} \\
\texttt{<NonZeroInt>} &::= \texttt{<NonZeroDigit>} \mid \texttt{<NonZeroInt>} \mathbin{+\!\!+} \texttt{<Digit>} \\
\texttt{<NonZeroDigit>} &::= 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9 \\
\texttt{<Digit>} &::= 0 \mid \texttt{<NonZeroDigit>} \\
\texttt{<Exp>} &::= \texttt{<MultDivExp>} \mid \texttt{<Exp> <AddSubOp> <MultDivExp>} \\
\texttt{<AddSubOp>} &::= + \mid - \\
\texttt{<MultDivExp>} &::= \texttt{<Factor>} \mid \texttt{<MultDivExp> <MultDivOp> <Factor>} \\
\texttt{<Factor>} &::= \texttt{<Int>} \mid \texttt{<Var>} \mid \texttt{(<Exp>)} \\
\texttt{<MultDivOp>} &::= * \mid / \\
\texttt{<Body>} &::= \texttt{<StmtList>} \mid \texttt{<Return>} \mid \texttt{<StmtList>} \mathbin{+\!\!+} \texttt{<Return>} \\
\texttt{<StmtList>} &::= \texttt{<Statement>} \mid \texttt{<StmtList>} \mathbin{+\!\!+} \texttt{<Statement>} \\
\texttt{<Statement>} &::= \texttt{<Declaration>;} \mid \texttt{<Assignment>;} \\
\texttt{<Declaration>} &::= \texttt{var <Var>} \mid \texttt{var <Var> = <Exp>} \\
\texttt{<Assignment>} &::= \texttt{<Var> = <Exp>} \\
\texttt{<Return>} &::= \texttt{<Exp>} \\
\texttt{<Call>} &::= \texttt{<Var> = <Exp>} \\
\texttt{<Fn>} &::= \\
\texttt{<FnSig>} &::= \texttt{<Var>()} \\
\texttt{<Letter>} &::= [a - zA - Z] \\
\texttt{<Var>} &::= \texttt{<Letter>} \mid \texttt{<Var>} \mathbin{+\!\!+} \texttt{<Letter>} \mid \texttt{<Var>} \mathbin{+\!\!+} \texttt{<Digit>}
\end{aligned}
$$